

Selected Solutions to *Linear Algebra Done Wrong*

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Introduction

Linear Algebra Done Wrong by Prof. Sergei Treil is a well-known linear algebra reference book for college students. However, I found no solution manual was available when I read this book and did the exercises. In contrast, the solution manual for the other famous book *Linear Algebra Done Right* is readily available for its greater popularity (without any doubt, *Dong Wrong* is also an excellent book). Reference solutions are important since insights tend to hide behind mistakes and ambiguity, which may be missed if there is no good way to check our answers. They also help to save time when there is no hint or the hint is obscure.

After scanning all and doing most problems in the first seven chapters (2014 version), I share those valuable ones here (those are relatively hard or insightful from my perspective. I read this book for reviewing and deeper mathematical understanding). The rest problems should be tractable even for a novice in linear algebra. Chapter 7 has a few easy problems and none is selected herein. The last Chapter 8 deals with dual spaces and tensors, which could be advanced for most readers and is also not covered. As references aiming to facilitate readers' learning process, the correctness and optimality of the solutions are not guaranteed. Lastly, do *not* copy the contents as it can violate your college's academic rules.

I'm now a PhD student with limited time to craft the contents. Typos and grammar mistakes are inevitable. Available at huangjingonly@gmail.com if any feedback. Cheers.

Chapter 1. Basic Notations

1.4. Prove that a zero vector of a vector space V is unique.

Proof Suppose there exist two different zero vectors $\mathbf{0}_1$ and $\mathbf{0}_2$. Then, for any $\mathbf{v} \in V$

$$\mathbf{v} + \mathbf{0}_1 = \mathbf{v}$$

$$\mathbf{v} + \mathbf{0}_2 = \mathbf{v}.$$

Then,

$$\begin{aligned}\mathbf{0}_1 &= \mathbf{0}_1 + \mathbf{0}_2 \text{ (}\mathbf{0}_2 \text{ is a zero vector)} \\ &= \mathbf{0}_2 + \mathbf{0}_1 \text{ (commutativity)} \\ &= \mathbf{0}_2 \text{ (}\mathbf{0}_1 \text{ is a zero vector)}\end{aligned}$$

Thus, the zero vector is unique.

1.6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

Proof Assume there exist two different additive inverses $\mathbf{w}_1$ and $\mathbf{w}_2$ of vector $\mathbf{v} \in V$, namely

$$\begin{aligned}\mathbf{v} + \mathbf{w}_1 &= \mathbf{0} \\ \mathbf{v} + \mathbf{w}_2 &= \mathbf{0}.\end{aligned}$$

We have

$$\begin{aligned}\mathbf{w}_1 &= \mathbf{w}_1 + \mathbf{0} \\ &= \mathbf{w}_1 + (\mathbf{v} + \mathbf{w}_2) \text{ (}\mathbf{w}_2 \text{ is an additive inverse)} \\ &= (\mathbf{w}_1 + \mathbf{v}) + \mathbf{w}_2 \text{ (associativity)} \\ &= \mathbf{0} + \mathbf{w}_2 \text{ (}\mathbf{w}_1 \text{ is an additive inverse)} \\ &= \mathbf{w}_2.\end{aligned}$$

Therefore, the additive inverse is unique.

1.7. Prove that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v} \in V$.

Proof Note that

$$0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$$

Denote the additive inverse of $0\mathbf{v}$ as $-0\mathbf{v}$, i.e., $0\mathbf{v} + (-0\mathbf{v}) = \mathbf{0}$. Combining with the above equation, we have

$$\begin{aligned}0\mathbf{v} + (-0\mathbf{v}) &= 0\mathbf{v} + 0\mathbf{v} + (-0\mathbf{v}) \\ \mathbf{0} &= 0\mathbf{v} + (0\mathbf{v} + (-0\mathbf{v})) \\ \mathbf{0} &= 0\mathbf{v} + \mathbf{0} \\ \mathbf{0} &= 0\mathbf{v}\end{aligned}$$

Thus, $0\mathbf{v} = \mathbf{0}$ as desired.

1.8. Prove that for any vector $\mathbf{v}$ its additive inverse $-\mathbf{v}$ is given by $(-1)\mathbf{v}$.

Proof $\mathbf{v} + (-1)\mathbf{v} = (1 - 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$ and we know from Problem 1.6 that the additive inverse is unique. Hence, $-\mathbf{v} = (-1)\mathbf{v}$.

2.5. Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1}$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

Proof Take $\mathbf{v}_{r+1}$ that can not be represented as $\sum_{k=1}^r \alpha_k \mathbf{v}_k$. Such a $\mathbf{v}_{r+1}$ exists because $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are not generating. Now we need to show $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ are linearly independent. Suppose that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ are linearly dependent, i.e.,

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_r \mathbf{v}_r + \alpha_{r+1} \mathbf{v}_{r+1} = \mathbf{0},$$

and $\sum_{k=1}^{r+1} |\alpha_k| \neq 0$. If $\alpha_{r+1} = 0$, then

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_r \mathbf{v}_r = \mathbf{0},$$

and $\sum_{k=1}^r |\alpha_k| \neq 0$. This contradicts that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are linearly independent. So $\alpha_{r+1} \neq 0$ and $\mathbf{v}_{r+1}$ can be represented as

$$\mathbf{v}_{r+1} = -\frac{1}{\alpha_{r+1}} \sum_{k=1}^r \alpha_k \mathbf{v}_k.$$

This contradicts the premise that $\mathbf{v}_{r+1}$ can not be represented by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. Thus, the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

2.6. Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly *independent*?

Solution 1 (By algebraic computation) It is impossible that $\mathbf{w}_1, \mathbf{w}_2$ and $\mathbf{w}_3$ are linearly independent. Note that $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ can also be expressed by $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ as follows

$$\begin{cases} \mathbf{v}_1 &= \frac{\mathbf{w}_1 + \mathbf{w}_3 - \mathbf{w}_2}{2} \\ \mathbf{v}_2 &= \frac{\mathbf{w}_1 + \mathbf{w}_2 - \mathbf{w}_3}{2} \\ \mathbf{v}_3 &= \frac{\mathbf{w}_2 + \mathbf{w}_3 - \mathbf{w}_1}{2}. \end{cases}$$

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent. Without loss of generality, suppose $\mathbf{v}_3 = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2$. Then, $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$. Substituting $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ with $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ above yields

$$(\alpha_1 + \alpha_2 + 1)\mathbf{w}_1 + (\alpha_2 - \alpha_1 - 1)\mathbf{w}_2 + (\alpha_1 - \alpha_2 - 1)\mathbf{w}_3 = \mathbf{0}.$$

If $\mathbf{w}_1, \mathbf{w}_2$ and $\mathbf{w}_3$ are linearly independent, the equation only admits a trivial combination, namely,

$$\begin{cases} \alpha_1 + \alpha_2 + 1 = 0 \\ \alpha_2 - \alpha_1 - 1 = 0 \\ \alpha_1 - \alpha_2 - 1 = 0 \end{cases}$$

which, however, does not have a solution for α_1, α_2 . Thus, a non-trivial combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ equals $\mathbf{0}$ and $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are linearly dependent.

Solution 2 (See Problem 5.4) From the perspective of dimensions of spanning systems, note that $3 > \dim \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) \geq \dim \text{span}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3) = \text{number of linearly independent vectors in } \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.

3.2. Let a linear transformation in $\mathbb{R}^2$ be the reflection in the line $x_1 = x_2$. Find its matrix.

Solution 1. Reflection is a linear transformation. It is completely defined on the standard basis. And $\mathbf{e}_1 = [1 \ 0]^T \xrightarrow{T} \mathbf{r}_1 = [0 \ 1]^T$, $\mathbf{e}_2 = [0 \ 1]^T \xrightarrow{T} \mathbf{r}_2 = [1 \ 0]^T$. The matrix is the combination of the two transformed standard basis as its first and second columns, i.e.,

$$T = [\mathbf{r}_1 \ \mathbf{r}_2] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Solution 2. (A more general method) Transformations such as reflection, rotation w.r.t. coordinate axes are easy to express. The general idea is to convert transformations into forms w.r.t. coordinate axes (also see the example on Page 20 in the book).

Let α be the angle between the x -axis and the reflection line. The reflection can be achieved through the following steps: First, rotate the line around the origin $-\alpha$ so that the line aligns with the x -axis. $x_1 = x_2$ passes through the origin. If the reflection line does not pass through the origin, translation is needed to make it pass through the origin. (*Homogeneous coordinates* will be needed since translation is not a linear transformation if represented in standard coordinates). Secondly, perform reflection about the x -axis, whose matrix is easy to get. Lastly, rotate the current frame back to its original location or perform other corresponding inverse transformations. In this problem,

$$T = R_z(-\alpha) \cdot Ref \cdot R_z(\alpha).$$

That is

$$\begin{aligned} T &= \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) \\ \sin(-\alpha) & \cos(-\alpha) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \\ &= \begin{bmatrix} \cos(-\frac{\pi}{4}) & -\sin(-\frac{\pi}{4}) \\ \sin(-\frac{\pi}{4}) & \cos(-\frac{\pi}{4}) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(\frac{\pi}{4}) & -\sin(\frac{\pi}{4}) \\ \sin(\frac{\pi}{4}) & \cos(\frac{\pi}{4}) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

3.7. Show that any linear transformation in $\mathbb{C}$ (treated as a complex vector space) is a multiplication by $\alpha \in \mathbb{C}$.

Proof Suppose a linear transformation is $T : \mathbb{C} \Rightarrow \mathbb{C}$ and $T(1) = a + ib$ for

two real numbers a and b . Then, $T(-1) = -T(1) = -a - ib$. Note that $i^2 = -1$. $T(-1) = T(i^2) = iT(i)$. Thus

$$T(i) = \frac{-a - ib}{i} = i(a + ib).$$

For any $\omega = x + iy \in \mathbb{C}, x, y \in \mathbb{R}$,

$$\begin{aligned} T(\omega) &= T(x + iy) = xT(1) + yT(i) \\ &= x(a + ib) + yi(a + ib) \\ &= (x + iy)(a + ib) \\ &= \omega T(1) \\ &= \omega \alpha \end{aligned}$$

where $\alpha = T(1)$.

5.4. Find the matrix of the orthogonal projection in $\mathbb{R}^2$ onto the line $x_1 = -2x_2$.

Solution (Directly projecting the standard basis onto the line also works and mainly involves some trigonometry computations.) Following similar steps presented in Problem 3.2, we have

$$\begin{aligned} T &= R(\alpha)PR(-\alpha) \\ &= R(\alpha) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} R(-\alpha). \end{aligned}$$

$\alpha = \tan^{-1}(-\frac{1}{2})$, and the resulting matrix is

$$T = \begin{bmatrix} \frac{4}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix}.$$

5.6. Prove Theorem 5.1, i.e., prove that $\text{trace}(AB) = \text{trace}(BA)$.

Proof 1. (By computation)

$$\begin{aligned} \text{trace}(A_{m \times n} B_{n \times m}) &= \sum_{i=1}^m (AB)_{i,i} = \sum_{i=1}^m \left(\sum_{j=1}^n a_{i,j} b_{j,i} \right) = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} b_{j,i} \\ \text{trace}(B_{n \times m} A_{m \times n}) &= \sum_{i=1}^n (BA)_{i,i} = \sum_{i=1}^n \left(\sum_{j=1}^m b_{i,j} a_{j,i} \right) = \sum_{i=1}^n \sum_{j=1}^m b_{i,j} a_{j,i} \\ &= \sum_{i=1}^n \sum_{j=1}^m a_{j,i} b_{i,j} = \sum_{j=1}^m \sum_{i=1}^n a_{j,i} b_{i,j} \text{ (commutativity of } \sum) \\ &= \text{trace}(A_{m \times n} B_{n \times m}) \end{aligned}$$

Proof 2. (By linear transformation) As advised on Page 23, another approach to proving this theorem is from a linear transformation perspective, which saves many algebraic manipulations. Consider two transformations T and T_1 , both acting from $M_{n \times m}$ to $\mathbb{F}^1$ ($\mathbb{F}$ can be $\mathbb{R}$ or $\mathbb{C}$) defined by

$$T(X) = \text{trace}(AX) \quad T_1(X) = \text{trace}(XA) \quad A \in \mathbb{F}^{m \times n}, X \in \mathbb{F}^{n \times m}.$$

Clearly, T and T_1 are linear transformations as both matrix multiplication and $\text{trace}()$ are linear transformations. Next, we show $T = T_1$, for which we need to show $T = T_1$ on a basis of $\mathbb{F}^{n \times m}$. Consider the basis $E_{i,j} \in \mathbb{F}^{n \times m}$ ($1 \leq i \leq n, 1 \leq j \leq m$) with its entry on the i -th row and j -th column being 1 and 0 elsewhere. Then

$$\begin{aligned} T(E_{i,j}) &= \text{trace}(AE_{i,j}) = \text{trace}([\mathbf{0}_{m \times 1} \quad \mathbf{0}_{m \times 1} \cdots \underbrace{\mathbf{a}_i}_{j\text{-th col}} \cdots \mathbf{0}_{m \times 1}]_{m \times m}) = (\mathbf{a}_i)_{[j]} \\ &= A_{[j,i]} \end{aligned}$$

where $\mathbf{a}_i$ is the i -th column vector of A . $[\cdot]$ represents the indexing operation.

Analogously, for T_1

$$T_1(E_{i,j}) = \text{trace}(E_{i,j}A) = \text{trace}\left(\begin{bmatrix} \mathbf{0}_{n \times 1} \\ \mathbf{0}_{n \times 1} \\ \cdots \\ \underbrace{\hat{\mathbf{a}}_j}_{i\text{-th row}} \\ \cdots \\ \mathbf{0}_{n \times 1} \end{bmatrix}_{n \times n}\right) = (\hat{\mathbf{a}}_j)_{[i]} = A_{[j,i]}$$

where $\hat{\mathbf{a}}_j$ represents the j -th row of A . Here it will be more convenient to calculate $E_{i,j}A$ using the row by coordinate rule. Thus, we have $T(E_{i,j}) = T_1(E_{i,j})$. T and T_1 are the same on a basis, so they are equal. Thus, for any $B \in \mathbb{F}^{n \times m}$, $T(B) = \text{trace}(AB) = T_1(B) = \text{trace}(BA)$.

5.7. Find the matrix of the reflection through the line $y = -2x/3$. Perform all the multiplications.

Solution As above,

$$\begin{aligned} T &= R(\alpha)R_{ref}R(-\alpha) \\ &= R(\alpha) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} R(-\alpha). \end{aligned}$$

$$\alpha = \tan^{-1}\left(-\frac{2}{3}\right),$$

$$T = \begin{bmatrix} \frac{5}{13} & -\frac{12}{13} \\ -\frac{12}{13} & \frac{5}{13} \end{bmatrix}.$$

6.1. Prove that if $A : V \rightarrow W$ is an isomorphism (i.e. an invertible linear transformation) and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V , then $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ is a basis in W .

Proof For any $\mathbf{w} \in W$, $A^{-1}\mathbf{w} = \mathbf{v} \in V$,

$$\begin{aligned}\mathbf{w} &= A\mathbf{v} = A[\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n][v_1 \ v_2 \ \dots \ v_n]^\top \\ &= [A\mathbf{v}_1 \ A\mathbf{v}_2 \ \dots \ A\mathbf{v}_n][v_1 \ v_2 \ \dots \ v_n]^\top.\end{aligned}$$

Thus $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ is generating in W . Next we only need to show that $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ are linearly independent. If they are not linearly independent, without loss of generality, suppose $A\mathbf{v}_1$ can be expressed as a linear combination of $A\mathbf{v}_2 \ A\mathbf{v}_3 \ \dots \ A\mathbf{v}_n$. Multiplying them with A^{-1} in the left side, it results in that $\mathbf{v}_1$ can be expressed by $\mathbf{v}_2 \ \dots \ \mathbf{v}_n$, which contradicts the fact that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V . The proposition is proved.

7.4. Let $\mathbf{X}$ and $\mathbf{Y}$ be subspaces of a vector space $\mathbf{V}$. Using the previous exercise, show that $\mathbf{X} \cup \mathbf{Y}$ is a subspace if and only if $\mathbf{X} \subset \mathbf{Y}$ or $\mathbf{Y} \subset \mathbf{X}$.

Proof The sufficiency is obvious and easy to verify. For the necessity, suppose $\mathbf{X} \not\subset \mathbf{Y}$, $\mathbf{Y} \not\subset \mathbf{X}$, and $\mathbf{X} \cup \mathbf{Y}$ is a subspace of $\mathbf{V}$. Then there are vectors $\mathbf{x} \in \mathbf{X}$, $\mathbf{y} \in \mathbf{Y}$ and $\mathbf{x} \notin \mathbf{Y}$, $\mathbf{y} \notin \mathbf{X}$. According to Problem 7.3, $\mathbf{x} + \mathbf{y} \notin \mathbf{X}$, $\mathbf{x} + \mathbf{y} \notin \mathbf{Y}$. As a result, $\mathbf{x} + \mathbf{y} \notin \mathbf{X} \cup \mathbf{Y}$, i.e., $\mathbf{x} \in \mathbf{X} \cup \mathbf{Y}$, $\mathbf{y} \in \mathbf{X} \cup \mathbf{Y}$, but $\mathbf{x} + \mathbf{y} \notin \mathbf{X} \cup \mathbf{Y}$, which contradicts that $\mathbf{X} \cup \mathbf{Y}$ is a subspace. Thus, $\mathbf{X} \subset \mathbf{Y}$ or $\mathbf{Y} \subset \mathbf{X}$.

8.5. A transformation T in $\mathbb{R}^3$ is a rotation about the line $y = x + 3$ in the x - y plane through an angle γ . Write a 4×4 matrix corresponding to this transformation.

You can leave the result as a product of matrices.

Solution For a general spatial rotation about a given line that passes through the origin through an angle γ , the 3×3 rotation matrix can be attained by

$$R = R_x^{-1} R_y^{-1} R_z(\gamma) R_y R_x$$

where the rotation by γ is appointed to be performed around z -axis. R_x and R_y are rotations used to align the original line direction with z -axis which can be determined by simple trigonometry.

For this problem, the line $y = x + 3$ does not go through the origin, so an extra step T_0 is needed to translate the line to make it pass the origin and homogeneous coordinates are applied:

$$R_{4 \times 4} = T_0^{-1} R_x^{-1} R_y^{-1} R_z(\gamma) R_y R_x T_0$$

where rotation matrices are also in their 4×4 forms. T_0 is not unique for the translation to make two parallel lines align. For example, consider the

following matrix:

$$T_0 = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Move } y = x + 3 \text{ to pass through the origin as } y = x.)$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Rotate } y = x \text{ about the } x\text{-axis } \pi/2.)$$

$$R_y = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Rotate the line about the } y\text{-axis } -\pi/4.)$$

$$R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 & 0 \\ \sin \gamma & \cos \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Rotate about the } z\text{-axis } \gamma.)$$

Combining all the matrices above and their inverses yields

$$R_{4 \times 4} = \begin{bmatrix} \frac{\cos \gamma + 1}{2} & -\frac{1 - \cos \gamma}{2} & \frac{\sqrt{2} \sin \gamma}{2} & \frac{3 \cos \gamma - 3}{2} \\ \frac{1 - \cos \gamma}{2} & \frac{\cos \gamma + 1}{2} & -\frac{\sqrt{2} \sin \gamma}{2} & \frac{3 - 3 \cos \gamma}{2} \\ -\frac{\sqrt{2} \sin \gamma}{2} & \frac{\sqrt{2} \sin \gamma}{2} & \cos \gamma & -\frac{3\sqrt{2} \sin \gamma}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Note that the processing above showcases a general scenario where we need to perform two rotations about two coordinate axes in sequence to align the line with the third coordinate axis. In this problem, it is easier to align the line with the x -axis or y -axis since the line lies on the $x - y$ plane. For instance, conduct the rotation about the x -axis, $R_{4 \times 4}$ is given by

$$R_{4 \times 4} = T_0^{-1} R_z^{-1} R_x(\gamma) R_z T_0.$$

Specifically, T_0 remains the same and

$$R_z = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Rotate about the } z\text{-axis } -\pi/4.)$$

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma & 0 \\ 0 & \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ (Rotate about the } x\text{-axis } \gamma.)$$

After computation, it can be verified that this solution will also result in the identical $R_{4 \times 4}$ as shown above.

Chapter 2. Systems of Linear Equations

3.8. Show that if the equation $A\mathbf{x} = \mathbf{0}$ has unique solution (i.e. if echelon form of A has pivot in every column), then A is left invertible.

Proof $A\mathbf{x} = \mathbf{0}$ has a unique solution, then the solution is the trivial solution. The echelon form of A has a pivot in every column. Let the dimension of A be $m \times n$, then $m \geq n$. The row number is greater than or equal to the column number. The reduced echelon form of A can be denoted as

$$A_{re} = \begin{bmatrix} I_{n \times n} \\ \mathbf{0}_{(m-n) \times n} \end{bmatrix}.$$

Suppose A_{re} is obtained by a sequence of elementary row operation $E_1, E_2, \dots, E_k$,

$$A_{re} = E_k \dots E_2 E_1 A$$

E_i is $m \times m$. The left inverse of A is the first n rows of the product of E_i . i.e.

$$E_{left} = I_{n \times m} E_k \dots E_2 E_1,$$

where

$$I_{n \times m} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots \end{bmatrix}_{n \times m},$$

is used to extract the $I_{n \times n}$ identity matrix in A_{re} . $E_{left}A = I_{n \times m}A_{re} = I_{n \times n}$, thus A is left invertible.

5.5. Let vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be a basis in V . Show that $\mathbf{u} + \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w}, \mathbf{w}$ is also a basis in V .

Solution For any vector $\mathbf{x} \in V$, suppose $\mathbf{x} = x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w}$. It is easy to figure out that $\mathbf{x} = x_1(\mathbf{u} + \mathbf{v} + \mathbf{w}) + (x_2 - x_1)(\mathbf{v} + \mathbf{w}) + (x_3 - x_2 - x_1)\mathbf{w}$. Clearly, $\mathbf{u} + \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w}, \mathbf{w}$ are linearly independent (check that only the trivial solution exists for their linear combination to be zero), can express any $\mathbf{x}$, and thus form a basis in V .

7.4. Prove that if $A : X \rightarrow Y$ and V is a subspace of X then $\dim AV \leq$

rank A . (AV here means the subspace V transformed by the transformation A , i.e., any vector in AV can be represented as $A\mathbf{v}$, $\mathbf{v} \in V$). Deduce from here that $\text{rank}(AB) \leq \text{rank} A$.

Proof $\dim AV \leq \dim AX \leq \dim \text{Ran} A = \text{rank} A$

$\text{rank}(AB) = \dim \text{Ran} AB \leq \dim \text{Ran} A = \text{rank} A$ as $\text{Ran} AB \subset \text{Ran} A$. Thus, $\text{rank}(AB) \leq \text{rank} A$.

7.5. Prove that if $A : X \rightarrow Y$ and V is a subspace of X then $\dim AV \leq \dim V$. Deduce from here that $\text{rank}(AB) \leq \text{rank} B$.

Proof Suppose that $\dim V = k$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be a basis of V . The vector space AV can be spanned by: $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k$, which contains a basis of AV . $\dim AV = \text{number of linearly independent vectors in } A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k = \text{rank}[A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k] \leq k = \dim V$.

Similarly, assume $B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n]$, $\text{Rank} B = k \leq n$, and $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$ are linearly independent column vectors in B . $\text{rank}(AB) = \dim \text{Ran} AB = \dim \text{Ran}[A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_n] = \dim \text{Ran}[A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_k] = \text{rank}[A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_k] \leq k = \text{rank} B$. Thus, $\text{rank}(AB) \leq \text{rank} B$.

7.6. Prove that if the product AB of two $n \times n$ matrices is invertible, then both A and B are invertible. Do not use determinant for this problem.

Proof AB is invertible, $\text{rank}(AB) = n$. From Problem 7.5, we have $\text{rank}(AB) = n \leq \text{rank}(A) \leq n$. Thus $\text{rank} A = n$. From Problem 7.6, we have $\text{rank}(AB) = n \leq \text{rank}(B) \leq n$. Thus $\text{rank} B = n$. Both A and B have full rank and are invertible.

7.7. Prove that if $A\mathbf{x} = \mathbf{0}$ has unique solution, then the equation $A^T \mathbf{x} = \mathbf{b}$ has a solution for every right side $\mathbf{b}$. (*Hint*: count pivots)

Proof Suppose $A \in \mathbb{R}^{m \times n}$. Note that for $A\mathbf{x} = \mathbf{0}$, there is always a trivial solution $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$. We now know the trivial solution is unique, which indicates that the echelon form of A has a pivot at every column. $\text{Rank} A = \text{Rank} A^T = n$. Accordingly, the echelon form of A^T will have a pivot at every row since $A^T \in \mathbb{R}^{n \times m}$ has n rows. As a result, $A^T \mathbf{x} = \mathbf{b}$ is consistent for any $\mathbf{b}$.

7.14. Is it possible for a real matrix A that $\text{Ran} A = \text{Ker} A^T$? Is it possible for a complex A ?

Solution It is not possible for a real matrix, but possible for a complex one. First consider A is a real matrix. Suppose A is $m \times n$ and $\text{Ran} A = \text{Ker} A^T$. Then $\text{Ran} A \subset \text{Ker} A^T$, i.e., $A^T A \mathbf{v} = \mathbf{0}$ for any $\mathbf{v} \in \mathbb{R}^n$. This holds only when $A^T A = 0_{n \times n}$. Thus $A = 0_{m \times n}$. (By checking the diagonal entries of $A^T A$ equal to 0, it leads to the conclusion that the column vectors are all-zero vectors, e.g., $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$ with $\mathbf{a}_i \in \mathbb{R}^m$. $A^T A_{1,1} = \mathbf{a}_1^T \mathbf{a}_1 = \|\mathbf{a}_1\|^2 = 0$, then $\mathbf{a}_1 = \mathbf{0}_{m \times 1}$.)

On the other hand, if $\text{Ran} A = \text{Ker} A^T$, $\text{Ker} A^T \subset \text{Ran} A$, i.e., if

$A^T \mathbf{b} = \mathbf{0}$, then $A\mathbf{x} = \mathbf{b}$ has a solution. We already have $A = 0_{m \times n}$. For any $\mathbf{b} \in \mathbb{R}^m$, $A^T \mathbf{b} = \mathbf{0}$, i.e., $\text{Ker } A^T = \mathbb{R}^m$. But $\text{Ran } A = \{\mathbf{0}_m\}$, for $\mathbf{b} \neq \mathbf{0}$, $A\mathbf{x} = \mathbf{b}$ does not have a solution. This is contradictory. So it is not possible for a real A that $\text{Ran } A = \text{Ker } A^T$.

It is possible for a complex matrix A that $\text{Ran } A = \text{Ker } A^T$. A specific example is $A = \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}$ in Problem 7.13. When A is complex, $A^T A = 0_{n \times n} \nRightarrow A = 0_{m \times n}$. Particularly, $A^T A_{1,1} = \mathbf{a}_1^T \mathbf{a}_1 = a_{1,1}^2 + \dots + a_{1,m}^2 = 0 \neq |a_{1,1}|^2 + \dots + |a_{1,m}|^2$ since $a_{1,i}$ is a complex number. $\mathbf{a}_1$ may not be a zero vector, e.g., $\mathbf{a}_1 = [1, i]^T$.

8.5. Prove that if A and B are similar matrices then $\text{trace } A = \text{trace } B$. (*Hint*: recall how $\text{trace}(XY)$ and $\text{trace}(YX)$ are related.)

Proof $\text{trace}(A) = \text{trace}(Q^{-1}BQ) = \text{trace}(Q^{-1}QB) = \text{trace}(B)$. (Note that $\text{trace}(AB) = \text{trace}(BA)$ as long as AB, BA can be performed.)

Chapter 3. Determinants

3.4. A square matrix ($n \times n$) is called skew-symmetric (or antisymmetric) if $A^T = -A$. Prove that if A is skew-symmetric and n is odd, then $\det A = 0$. Is this true for even n ?

Proof $\det A = \det A^T = \det(-A) = (-1)^n \det A$ by using the properties of determinant and skew-symmetric matrices. If n is odd, $(-1)^n = -1$, we have $\det A = -\det A$, thus $\det A = 0$.

If n is even, we just have $\det A = \det A$, so the result above will generally not hold.

3.5. A square matrix is called *nilpotent* if $A^k = \mathbf{0}$ for some positive integer k . Show that for a nilpotent matrix A , $\det A = 0$.

Proof $\det A^k = (\det A)^k = \det \mathbf{0} = 0$, thus $\det A = 0$.

3.6. Prove that if A and B are similar, then $\det A = \det B$.

proof A and B are similar, then $A = Q^{-1}BQ$ for an invertible matrix Q .

$$\begin{aligned} \det A &= \det Q^{-1}BQ \\ &= (\det Q^{-1})(\det B)(\det Q) \\ &= (\det Q^{-1})(\det Q)(\det B) \\ &= (\det Q^{-1}Q)(\det B) \\ &= (\det I)(\det B) \\ &= \det B. \end{aligned}$$

3.7. A real square matrix Q is called orthogonal if $Q^T Q = I$. Prove that if Q is an orthogonal matrix then $\det Q = \pm 1$.

Proof $\det Q^T Q = (\det Q^T)(\det Q) = (\det Q)^2 = \det I = 1$. $\det Q = \pm 1$.

3.9. Let points A, B and C in the plane $\mathbb{R}^2$ have coordinates (x_1, y_1) , (x_2, y_2) and (x_3, y_3) respectively. Show that the area of triangle ABC is the absolute value of

$$\frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

Hint: use row operation and geometric interpretation of 2×2 determinants (area).

Proof The area of the triangle $\triangle ABC$ is half of the parallelogram defined by neighboring sides AB, AC , which also can be computed by

$$\begin{aligned} S_{\triangle ABC} &= \frac{1}{2} \text{abs} \left(\begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix} \right) \\ &= \frac{1}{2} |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)|. \end{aligned}$$

In the same time, if we use row reduction to check the determinant

$$\begin{aligned} \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix} &= \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 0 & x_2 - x_1 & y_2 - y_1 \\ 0 & x_3 - x_1 & y_3 - y_1 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 0 & x_2 - x_1 & y_2 - y_1 \\ 0 & 0 & y_3 - y_1 - (y_2 - y_1) \frac{x_3 - x_1}{x_2 - x_1} \end{vmatrix} \\ &= \frac{1}{2} ((x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)). \end{aligned}$$

We assume that $x_2 - x_1 \neq 0$ and it can be verified if $x_2 - x_1 = 0$, the result still holds. Considering the absolute value, the conclusion holds.

3.10. Let A be a square matrix. Show that block triangular matrices

$$\begin{bmatrix} I & * \\ \mathbf{0} & A \end{bmatrix} \quad \begin{bmatrix} A & * \\ \mathbf{0} & I \end{bmatrix} \quad \begin{bmatrix} I & \mathbf{0} \\ * & A \end{bmatrix} \quad \begin{bmatrix} A & \mathbf{0} \\ * & I \end{bmatrix}$$

all have determinant equal to $\det A$. Here $*$ can be anything.

Proof Considering performing row reduction to make A triangular, the whole matrix will be triangular and the rest part on the diagonal remains I . Thus the determinant of the block matrix equals to $\det A$.

(Problem 3.11 and 3.12 are just applications of the conclusion of Problem 3.10. The hint just tells the answer.)

4.2. Let P be a permutation matrix, i.e., an $n \times n$ matrix consisting of

zeros and ones and such that there is exactly one 1 in every row and every column.

- a) Can you describe the corresponding linear transformation? That will explain the name.
- b) Show that P is invertible. Can you describe P^{-1} ?
- c) Show that for some $N > 0$

$$P^N := \underbrace{PP \dots P}_{N \text{ times}} = I.$$

Use the fact that there are only finitely many permutations.

Solution a) Consider the linear transformation $\mathbf{y} = P\mathbf{x}$ and rows of P . There is only one 1 in each row of P . Suppose in the first row of P , $P_{1,j} = 1$, then $y_1 = \mathbf{p}_1\mathbf{x} = x_j$, where $\mathbf{p}_1$ is the first row of P . Namely, x_j is moved to the 1st place after the linear transformation. Similarly, for the second row of P , suppose $P_{2,k} = 1$, then $y_2 = x_k$ and x_k is moved to the second place, so on and so forth. There is also only one 1 in each column, then the column indices in 1 entries $P_{1,j}, P_{2,k}, P_{3,m}, \dots$ comprise a permutation of n as $(j, k, m, \dots)$. This guarantees there is no repeated entries in $\mathbf{y}$ ($y_i \neq y_j$ if $i \neq j$). After multiplying by the permutation matrix P , the elements in $\mathbf{x}$ change their orders to $[x_j, x_k, x_m, \dots]^T$. (Considering from the perspective of column vectors of P also works.)

b) Suppose P is invertible, by multiplying P^{-1} , $\mathbf{x} = P^{-1}\mathbf{y}$. But we know $y_1 = x_j$, then we have $P_{j,1}^{-1} = 1$ so that x_j can return to its original position. Similarly, $y_2 = x_k$, then $P_{k,2}^{-1} = 1$. Following this, we can see that $P_{i,j}^{-1} = P_{j,i}$. So P is invertible and $P^{-1} = P^T$.

c) Note that $P\mathbf{x}, P^2\mathbf{x}, P^3\mathbf{x} \dots P^N\mathbf{x}$ are all permutations of $(x_1, x_2, \dots, x_n)$. If P^N can never equal to I , $P\mathbf{x}, P^2\mathbf{x}, P^3\mathbf{x} \dots P^N\mathbf{x}$ will be different permutations. And N can be infinitely big, so there will be infinitely many permutations of $(x_1, x_2, \dots, x_n)$, which is impossible. Thus there must be some $N > 0$, $P^N = I$. In fact, there are $n!$ different permutations of n distinct elements, $N \leq n!$.

Exercises Part 5 and Part 7 in this chapter are not difficult. Some ideas and answers for reference:

- Problem 5.3, we can use the last column cofactor expansion and the left matrix $(A + tI)_{i,j}$ is a triangular matrix. The final expression is $\det(A + tI) = a_0 + a_1t + a_2t^2 + \dots + a_{n-1}t^{n-1}$. The order of -1 in each term is even.
- Problem 5.7, more than $n!$ multiplications is needed. We can use induction to prove it. Let $f(n)$ be the multiplication number to compute

the determinant of an $n \times n$ matrix. Then the induction formula is $f(n+1) = (n+1)(f(n)+1)$ with $f(1) = 0$ (the computation of the sign term $(-1)^{i+j}$ is ignored.) and the general formula is $f(n) = n! \sum_{i=1}^{n-1} \frac{1}{i!}$. See this StackExchange page.

- Problem 7.4 and Problem 7.5, consider $\det RA = (\det R)(\det A) = \det A$, where R is a rotation matrix with its determinant equal to 1. For proof of the parallelogram area, we can also utilize the parameter angle, i.e., $\mathbf{v}_1 = [x_1, y_1]^T = [v_1 \cos \alpha, v_1 \sin \alpha]^T$, $\mathbf{v}_2 = [x_2, y_2]^T = [v_2 \cos \beta, v_2 \sin \beta]^T$. v_1, v_2 are the lengths of $\mathbf{v}_1, \mathbf{v}_2$, respectively. α, β represent angles between vectors and the x -axis positive direction. Then

$$\begin{aligned} \det A &= \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1 \\ &= v_1 v_2 (\cos \alpha \sin \beta - \cos \beta \sin \alpha) \\ &= v_1 v_2 \sin(\beta - \alpha). \end{aligned}$$

$\beta - \alpha$ is the angle from $\mathbf{v}_1$ to $\mathbf{v}_2$.

Chapter 4. Introduction to Spectral Theory (Eigenvalues and Eigenvectors)

1.1. (Part) True or false:

- b) If a matrix has one eigenvector, it has infinitely many eigenvectors;
True, if $A\mathbf{x} = \lambda\mathbf{x}$, $A(\alpha\mathbf{x}) = \lambda(\alpha\mathbf{x})$, α is an arbitrary scalar but zero. $\alpha\mathbf{x}$ is also an eigenvector of A .
- c) There exists a square matrix with no real eigenvalues;
True, e.g., a 2D rotation matrix $R_\alpha, \alpha \neq n\pi$.
- d) There exists a square matrix with no (complex) eigenvectors;
False, when discussing in complex space, there are always eigenvalues and as a result, $A - \lambda I$ has a null space containing elements beyond the zero vector.
- f) Similar matrices always have the same eigenvectors;
False, if A, B are similar and $A = SBS^{-1}$. If $A\mathbf{x} = \lambda\mathbf{x}$, then $SBS^{-1}\mathbf{x} = \lambda\mathbf{x}$. i.e., $B(S^{-1}\mathbf{x}) = \lambda(S^{-1}\mathbf{x})$, $S^{-1}\mathbf{x}$ is an eigenvector of B , not $\mathbf{x}$.
- g) The sum of two eigenvectors of a matrix A is always an eigenvector;
False, only if the two eigenvectors correspond to the same eigenvalue.

1.6. An operator A is called *nilpotent* if $A^K = \mathbf{0}$ for some K . Prove that if A is nilpotent, then $\sigma(A) = \{0\}$ (i.e., that 0 is the only eigenvalue of A).

Proof Note that if λ is a nonzero eigenvalue of A and $A\mathbf{x} = \lambda\mathbf{x}$. Then $A^2\mathbf{x} = A(\lambda\mathbf{x}) = \lambda^2\mathbf{x}$, $A^3\mathbf{x} = A(\lambda^2\mathbf{x}) = \lambda^3\mathbf{x}$... $A^k\mathbf{x} = \lambda^k\mathbf{x}$. In other words, if $\lambda \in \sigma(A)$, $\lambda^k \in \sigma(A^k)$. Now $A^k = \mathbf{0}$, $\sigma(A^k) = \{0\}$. Then 0 is the only eigenvalue of A .

1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in a vector space V . Assume also that the first k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ of the basis are eigenvectors of an operator A , corresponding to an eigenvalue λ (i.e. that $A\mathbf{v}_j = \lambda\mathbf{v}_j, j = 1, 2, \dots, k$). Show that in this basis the matrix of the operator A has block triangular form

$$\begin{bmatrix} \lambda I_k & * \\ \mathbf{0} & B \end{bmatrix}$$

where I_k is $k \times k$ identity matrix and B is some $(n - k) \times (n - k)$ matrix.

Proof Note that relations $A\mathbf{v}_j = \lambda\mathbf{v}_j, j = 1, 2, \dots, k$ (A is an abstract operator here, not a matrix) hold in the standard basis S , i.e., $A_{SS}\mathbf{v}_j = \lambda\mathbf{v}_j$ in matrix form. Denote the basis of $\mathbf{v}_1, \dots, \mathbf{v}_n$ as V . $A_{VV} = [I]_{VS}A_{SS}[I]_{SV}$ where $[I]$ is the coordinate change matrix and $[I]_{SV} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n]$. $[I]_{SV}A_{VV} = A_{SS}[I]_{SV} = A_{SS}[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = [\lambda\mathbf{v}_1, \lambda\mathbf{v}_2, \dots, \lambda\mathbf{v}_k, \dots, A_{SS}\mathbf{v}_n]$. Denote the i -th column of A_{VV} as $\mathbf{a}_i$. Consider $\mathbf{a}_1$, then $[I]_{SV}\mathbf{a}_1 = \sum_{i=1}^n a_{1,i}\mathbf{v}_i = \lambda\mathbf{v}_1$. Since $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis, then $\mathbf{a}_1$ can only be the form $\mathbf{a}_1 = [\lambda, 0, 0, \dots, 0]^T$. Similarly, check the first k columns of A_{VV} , $[I]_{SV}\mathbf{a}_i = \lambda\mathbf{v}_i$. Thus, A_{VV} has the block triangular form shown above.

Alternatively, a more simple way is to notice that the meaning of the i -th column of A_{VV} is the transformed result of the i -th base vector $\mathbf{v}_i$ expressed in the basis V . It is known that $A\mathbf{v}_j = \lambda\mathbf{v}_j, j = 1, 2, \dots, k$. For the first column of A_{VV} , $[\lambda\mathbf{v}_1]_V = [\lambda, 0, \dots, 0]^T = \mathbf{a}_1$. Similarly, checking the first k columns of A_{VV} leads to the triangular form.

1.9. Use the two previous exercises to prove that the geometric multiplicity of an eigenvalue cannot exceed its algebraic multiplicity.

Proof We consider the problem in the basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and A has the block triangular form shown in Problem 1.8. Note k is the number of linearly independent eigenvectors corresponding to λ_k , which is also the dimension of $\text{Ker}(A - \lambda_k I)$ (consider the equation $(A - \lambda_k I)\mathbf{x} = \mathbf{0}$). Namely, k is the geometric multiplicity of λ_k .

For the algebraic multiplicity, consider the determinant

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} (\lambda_k - \lambda)I_k & * \\ \mathbf{0} & B - \lambda I_{n-k} \end{vmatrix} \\ &= (\lambda_k - \lambda)^k \det(B - \lambda I_{n-k}). \end{aligned}$$

So the algebraic multiplicity of λ_k is at least k . It is further possible that λ_k is a root of the polynomial $\det(B - \lambda I_{n-k})$, then the algebraic multiplicity will just exceed k . Thus geometric multiplicity of an eigenvalue cannot

exceed its algebraic multiplicity.

1.10. Prove that determinant of a matrix A is the product of its eigenvalues (counting multiplicity).

Proof (Just use the hint) The characteristic polynomial of an $n \times n$ square matrix A is $\det(A - \lambda I)$ and we consider the roots of it in complex space. According to the fundamental theorem of algebra, $\det(A - \lambda I)$ has n roots counting multiplicity and can be factorized as $\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$. (Recall the formal definition of determinant, the highest order term of λ , λ^n , is generated by the diagonal product $\prod_{i=1}^n (a_{ii} - \lambda)$. Thus the sign of λ^n in the factorization is consistent with $\prod_{i=1}^n (a_{ii} - \lambda)$.) Let $\lambda = 0$, we will get $\det A = \lambda_1 \lambda_2 \dots \lambda_n$.

1.11. Prove that the trace of a matrix equals the sum of eigenvalues in three steps. First, compute the coefficient of λ^{n-1} in the right side of the equality

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda).$$

Then show that $\det(A - \lambda I)$ can be represented as

$$\det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda) \dots (a_{n,n} - \lambda) + q(\lambda).$$

where $q(\lambda)$ is polynomial of degree at most $n - 2$. And finally, comparing the coefficients of λ^{n-1} get the conclusion.

Proof First, recall the binomial theorem, the coefficient of λ^{n-1} in $\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$ is $C(\lambda^{n-1}) = (-1)^{n-1}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$. To get the term λ^{n-1} , we need to pick $-\lambda$ $n - 1$ times from the total n factors $\lambda_i - \lambda$. The last pick is λ_j from the factor whose $-\lambda$ is not picked. The resulting term is then $(-1)^{n-1} \lambda_j \lambda^{n-1}$. There are n combinations and the sum of each term is $C(\lambda^{n-1}) \lambda^{n-1}$.

Then, we show $\det(A - \lambda I)$ can be represented as

$$\det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda) \dots (a_{n,n} - \lambda) + q(\lambda).$$

That is to say in $\det(A - \lambda I)$, the term λ^{n-1} are all from $(a_{1,1} - \lambda)(a_{2,2} - \lambda) \dots (a_{n,n} - \lambda)$. This holds because λ only appears on the diagonal of $A - \lambda I$. Using the formal definition of determinant, if we pick $n - 1$ diagonal term with λ , then the last pick must also be on the diagonal. There is no other way to generate λ^{n-1} . Thus $q(\lambda)$ is a polynomial of degree at most $n - 2$. Then we know the coefficient of λ^{n-1} also equals to $C(\lambda^{n-1}) = (-1)^{n-1}(a_{1,1} + a_{2,2} + \dots + a_{n,n})$.

The coefficients derived in two ways are identical, so we have $\sum_{i=1}^n a_{i,i} = \sum_{i=1}^n \lambda_i$, namely, the trace a matrix equals the sum of eigenvalues.

2.1. Let A be $n \times n$ matrix. True or false:

- a) A^T has the same eigenvalues as A .
True, $\det(A - \lambda I) = \det(A - \lambda I)^T = \det(A^T - \lambda I)$
- b) A^T has the same eigenvectors as A .
False.
- c) If A is diagonalizable, then so is A^T .
True, $A = SDS^{-1}$, $A^T = (SDS^{-1})^T = (S^{-1})^T D^T S^T = (S^T)^{-1} D S^T$.

2.2. Let A be a square matrix with real entries, and let λ be its complex eigenvalue. Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$ is a corresponding eigenvector, $A\mathbf{v} = \lambda\mathbf{v}$. Prove that the $\bar{\lambda}$ is an eigenvalue of A and $A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$. Here $\bar{\mathbf{v}}$ is the complex conjugate of the vector $\mathbf{v}$, $\bar{\mathbf{v}} := (\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n)^T$.

Proof A is real matrix. Then $A\bar{\mathbf{v}} = A\bar{\mathbf{v}}$. In the same time, $\bar{A\mathbf{v}} = \overline{\lambda\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$. Thus $A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$.

Chapter 5. Inner Product Spaces

1.4. Prove that for vectors in an inner product space

$$\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\operatorname{Re}(\mathbf{x}, \mathbf{y}).$$

Recall that $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$.

Proof

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\|^2 &= (\mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x}, \mathbf{x} - \mathbf{y}) - (\mathbf{y}, \mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x}, \mathbf{x}) - (\mathbf{x}, \mathbf{y}) - (\mathbf{y}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - (\mathbf{x}, \mathbf{y}) - \overline{(\mathbf{x}, \mathbf{y})} \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\operatorname{Re}(\mathbf{x}, \mathbf{y}). \end{aligned}$$

Similarly, $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\operatorname{Re}(\mathbf{x}, \mathbf{y})$.

1.5. *Hint:* a) Check conjugate symmetry. b) Check linearity. c) Check conjugate symmetry.

1.7. Prove the parallelogram identity for an inner product space V ,

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2).$$

Proof

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) + (\mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x}, \mathbf{x}) + (\mathbf{x}, \mathbf{y}) + (\mathbf{y}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) + \\ &\quad (\mathbf{x}, \mathbf{x}) - (\mathbf{x}, \mathbf{y}) - (\mathbf{y}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) \\ &= 2(\mathbf{x}, \mathbf{x}) + 2(\mathbf{y}, \mathbf{y}) \\ &= 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2). \end{aligned}$$

- 1.8. Proof sketch:** a) Let $\mathbf{v} = \mathbf{x}$, then $(\mathbf{x}, \mathbf{x}) = 0$, $\mathbf{x} = \mathbf{0}$.
b) $(\mathbf{x}, \mathbf{v}_k) = 0, \forall k$, then $(\mathbf{x}, \mathbf{v}) = 0$, from conclusion in a), $\mathbf{x} = \mathbf{0}$.
c) $(\mathbf{x} - \mathbf{y}, \mathbf{v}_k) = 0, \forall k$, from b) $\mathbf{x} - \mathbf{y} = \mathbf{0}$, then $\mathbf{x} = \mathbf{y}$.

2.3. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be an orthonormal basis in V .

- a) Prove that for any $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n \alpha_k \bar{\beta}_k.$$

- b) Deduce from this *Parseval's identity*

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n (\mathbf{x}, \mathbf{v}_k) \overline{(\mathbf{y}, \mathbf{v}_k)}.$$

- c) Assume now that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is only an orthogonal basis, not an orthonormal one. Can you write down Parseval's identity in this case?

Proof

a)

$$(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n \alpha_k \mathbf{v}_k, \sum_{k=1}^n \beta_k \mathbf{v}_k \right) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \bar{\beta}_j (\mathbf{v}_i, \mathbf{v}_j) = \sum_{k=1}^n \alpha_k \bar{\beta}_k.$$

Because $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an orthonormal basis, $(\mathbf{v}_i, \mathbf{v}_j) = 0, i \neq j, (\mathbf{v}_i, \mathbf{v}_j) = 1, i = j$.

- b) Use $(\mathbf{x}, \mathbf{v}_k) = \alpha_k, (\mathbf{y}, \mathbf{v}_k) = \beta_k$ and conclusion in a).

- c) Use equation in a),

$$\begin{aligned} (\mathbf{x}, \mathbf{y}) &= \left(\sum_{k=1}^n \alpha_k \mathbf{v}_k, \sum_{k=1}^n \beta_k \mathbf{v}_k \right) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \bar{\beta}_j (\mathbf{v}_i, \mathbf{v}_j) = \sum_{k=1}^n \alpha_k \bar{\beta}_k (\mathbf{v}_k, \mathbf{v}_k) \\ &= \sum_{k=1}^n \alpha_k \bar{\beta}_k \|\mathbf{v}_k\|^2 = \sum_{k=1}^n \frac{(\mathbf{x}, \mathbf{v}_k) \overline{(\mathbf{y}, \mathbf{v}_k)}}{\|\mathbf{v}_k\|^2}. \end{aligned}$$

As the basis is only orthogonal, not orthonormal, then $(\mathbf{x}, \mathbf{v}_k) = (\alpha_k \mathbf{v}_k, \mathbf{v}_k) = \alpha_k \|\mathbf{v}_k\|^2$.

3.3 Complete an orthogonal system obtained in the previous problem to an orthogonal basis in $\mathbb{R}^3$, i.e., add to the system some vectors (how many?) to get an orthogonal basis.

Can you describe how to complete an orthogonal system to an orthogonal basis in general situation of $\mathbb{R}^n$ or $\mathbb{C}^n$?

Solution For 3D space, we already have 2 orthogonal vectors $\mathbf{v}_1, \mathbf{v}_2$ as the basis components. We just need another basis vector $\mathbf{v}_3$. The computation of $\mathbf{v}_3$ exploits the orthogonality, i.e., $(\mathbf{v}_1, \mathbf{v}_3) = 0, (\mathbf{v}_2, \mathbf{v}_3) = 0$. Expressed in matrix form, let $A = [\mathbf{v}_1, \mathbf{v}_2]$. Then solve $A^\top \mathbf{v}_3 = \mathbf{0}$. (Since it is in 3D space, using cross product is also simple.)

Generally, to complete an orthogonal system of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. Consider $A = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r]$, using the orthogonality, we compute the rest basis vectors by solving $A^\top \mathbf{v} = \mathbf{0}$, i.e., the rest basis vectors compose a basis of $\text{Ker } A^\top$ or $\text{Null } A^\top$. Then perform orthogonalization of $\text{Ker } A^\top$ basis. Then all the vectors make up an orthogonal basis.

3.9 (Using eigenvalues to compute determinants).

- Find the matrix of the orthogonal projection onto the one-dimensional subspace in $\mathbb{R}^n$ spanned by the vector $(1, 1, \dots, 1)^\top$;
- Let A be the $n \times n$ matrix with all entries equal 1. Compute its eigenvalues and their multiplicities (use the previous problem);
- Compute eigenvalues (and multiplicities) of the matrix $A - I$, i.e., of the matrix with zeros on the main diagonal and ones everywhere else;
- Compute $\det(A - I)$.

Solution a) Note that from Remark 3.5, we know $P_E = \sum_{k=1}^n \frac{1}{\|\mathbf{v}_k\|^2} \mathbf{v}_k \mathbf{v}_k^*$. For this one-dimensional subspace, it is

$$P_E = \frac{1}{n} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{n \times n}.$$

Alternatively, note that the i -th column of P_E is the orthogonal projection of the i -th standard basis vector $\mathbf{e}_i$ on the spanned space of $\mathbf{v} = (1, 1, \dots, 1)^\top$, which is $\frac{(\mathbf{e}_i, \mathbf{v})}{\|\mathbf{v}\|^2} \mathbf{v} = \frac{1}{n} \mathbf{v}$.

b) Note that $A = nP_E$. Suppose $A\mathbf{x} = \lambda\mathbf{x}$, then $nP_E\mathbf{x} = \lambda\mathbf{x}$. i.e., n times of the eigenvector's projection on the 1D subspace equals λ times of itself. It implies that the eigenvector's orthogonal projection is parallel to itself if $\lambda \neq 0$ or the orthogonal projection is $\mathbf{0}$ for $\lambda = 0$. Namely, there are two possibilities:

- The eigenvector is parallel to the basis of the 1D subspace $\mathbf{v}$, i.e., $\mathbf{x} = \alpha\mathbf{v}, \alpha \neq 0$. In this case, $P_E\mathbf{x} = \mathbf{x}$, then $\lambda = n$. The geometric multiplicity equals the subspace dimension, i.e., 1.
- The eigenvector is orthogonal to $\mathbf{v}$, i.e., $\mathbf{x} \perp \mathbf{v}$ and $P_E\mathbf{x} = \mathbf{0}, \lambda = 0$. We can find totally $n - 1$ linearly independent eigenvectors so the geometric multiplicity of eigenvalue $\lambda = 0$ is $n - 1$.

- c) $\det(A - I - \lambda I) = \det(A - (\lambda + 1)I)$. i.e., an eigenvalue of $A - I$ plus 1 is an eigenvalue of A . Then the eigenvalues of $A - I$ are equal to the eigenvalues of A minus 1. Thus the eigenvalues of $A - I$ are $n - 1$ with multiplicity 1 and -1 with multiplicity $n - 1$.
- d) $\det(A - I) = (n - 1)(-1)^{n-1}$, which equals $n - 1$ if n is odd, $1 - n$ if n is even.

3.10. (Legendre's polynomials) *Hint:* Using the Gram-Schmidt orthogonalization algorithm is sufficient. But remember to use the inner product defined in the problem, e.g., $\|1\|^2 = (1, 1) = \int_{-1}^1 1 \cdot 1 dt = 2$.

3.11. Let $P = P_E$ be the matrix of an orthogonal projection onto a subspace E . Show that

- The matrix P is *self-adjoint*, meaning that $P^* = P$.
- $P^2 = P$.

Remark The above 2 properties completely characterize orthogonal projection.

Proof a) From the orthogonality, we have $(\mathbf{x}, \mathbf{x} - P\mathbf{x}) = (\mathbf{x} - P\mathbf{x}, \mathbf{x}) = 0, \forall \mathbf{x}$.
 $(\mathbf{x}, \mathbf{x} - P\mathbf{x}) = (\mathbf{x} - P\mathbf{x})^* \mathbf{x} = (\mathbf{x}^* - \mathbf{x}^* P^*) \mathbf{x} = \mathbf{x}^* \mathbf{x} - \mathbf{x}^* P^* \mathbf{x} = 0$. On the other hand, $(\mathbf{x} - P\mathbf{x}, \mathbf{x}) = \mathbf{x}^* (\mathbf{x} - P\mathbf{x}) = \mathbf{x}^* \mathbf{x} - \mathbf{x}^* P\mathbf{x} = 0$. Subtract two equalities, $\mathbf{x}^* (P - P^*) \mathbf{x} = 0, \forall \mathbf{x}$. Then $P - P^* = 0_{n \times n}, P = P^*$.

b) Consider $(P\mathbf{x}, \mathbf{x} - P\mathbf{x}) = (\mathbf{x} - P\mathbf{x})^* P\mathbf{x} = (\mathbf{x}^* - \mathbf{x}^* P^*) P\mathbf{x} = \mathbf{x}^* (P - P^* P) \mathbf{x} = 0$. Thus $P = P^* P = P^2$ since $P = P^*$.

Intuitively, for any $\mathbf{x}$, $P^2 \mathbf{x} = P(P\mathbf{x}) = P\mathbf{x}$ since $P\mathbf{x} \in E$. The orthogonal projection of $P\mathbf{x}$ onto E equals itself. So $P^2 = P$.

3.13 Suppose P is the orthogonal projection onto a subspace E , and Q is the orthogonal projection onto the orthogonal complement $E^\perp$.

- What are $P + Q$ and PQ ?
- Show that $P - Q$ is its inverse.

Proof a) $P + Q = I$ since $(P + Q)\mathbf{x} = P\mathbf{x} + Q\mathbf{x} = P_E \mathbf{x} + Q_{E^\perp} \mathbf{x} = \mathbf{x}$.

$PQ = 0_{n \times n}$ as $\mathbf{x}^* PQ\mathbf{x} = \mathbf{x}^* P^* Q\mathbf{x} = (Q\mathbf{x}, P\mathbf{x}) = 0, \forall \mathbf{x}$ (using P is self-adjoint shown in Problem 3.11).

b) $(P - Q)^2 = (P - Q)(P - Q) = P^2 - PQ - QP + Q^2 = P^2 + Q^2 = P^2 + Q^2 + PQ + QP = (P + Q)^2 = I^2 = I$ (using $PQ = QP = 0$). i.e., $(P - Q)^{-1} = P - Q$.

4.5. *Minimal norm solution.* Let an equation $A\mathbf{x} = \mathbf{b}$ have a solution, and let A have non-trivial kernel (so the solution is not unique). Prove that

- There exists a unique solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{b}$ minimizing the norm $\|\mathbf{x}\|$, i.e., that there exists unique $\mathbf{x}_0$ such that $A\mathbf{x}_0 = \mathbf{b}$ and $\|\mathbf{x}_0\| \leq \|\mathbf{x}\|$ for any $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{b}$.

b) $\mathbf{x}_0 = P_{(\text{Ker } A)^\perp} \mathbf{x}$ for any $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{0}$.

Proof a) Suppose $\mathbf{x}_0, \mathbf{x}_1$ are solutions of $A\mathbf{x} = \mathbf{b}$. Then $A(\mathbf{x}_1 - \mathbf{x}_0) = \mathbf{0}$. i.e., $\mathbf{x}_1 - \mathbf{x}_0 \in \text{Ker } A$. As a result, $P_{(\text{Ker } A)^\perp}(\mathbf{x}_1 - \mathbf{x}_0) = \mathbf{0} = P_{(\text{Ker } A)^\perp} \mathbf{x}_1 - P_{(\text{Ker } A)^\perp} \mathbf{x}_0$. So we have $P_{(\text{Ker } A)^\perp} \mathbf{x}_1 = P_{(\text{Ker } A)^\perp} \mathbf{x}_0 = \text{const} := \mathbf{h}$.

Note that $\|\mathbf{x}\|^2 = \|P_{(\text{Ker } A)^\perp} \mathbf{x}\|^2 + \|\mathbf{x} - P_{(\text{Ker } A)^\perp} \mathbf{x}\|^2 \geq \|\mathbf{h}\|^2$ for any $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{b}$. When $\mathbf{x}_0 - P_{(\text{Ker } A)^\perp} \mathbf{x}_0 = \mathbf{0}$, $\mathbf{x}_0 = \mathbf{h}$, such a $\mathbf{x}_0$ has the smallest norm among all the solutions. The existence and uniqueness of $\mathbf{x}_0$ are guaranteed by $\mathbf{h}$.

b) It is shown above.

5.1. Show that for a square matrix A the equality $\det(A^*) = \overline{\det(A)}$ holds.

Proof $\det(A^*) = \det(\overline{A}^T) = \det(\overline{A}) = \overline{\det(A)}$.

5.3. Let A be an $m \times n$ matrix. Show that $\text{Ker } A = \text{Ker } (A^*A)$.

Proof It is easy to see $\text{Ker } A \subset \text{Ker } (A^*A)$. Next, we show $\text{Ker } (A^*A) \subset \text{Ker } A$. Consider $\|A\mathbf{x}\|^2 = (A\mathbf{x}, A\mathbf{x}) = \mathbf{x}^* A^* A \mathbf{x}$. Thus if $A^* A \mathbf{x} = \mathbf{0}$, we have $\mathbf{x}^* A^* A \mathbf{x} = \|A\mathbf{x}\|^2 = 0$, i.e., $A\mathbf{x} = \mathbf{0}$. Thus $\text{Ker } (A^*A) \subset \text{Ker } A$. As a result, we can conclude $\text{Ker } A = \text{Ker } (A^*A)$.

6.4. Show that a product of unitary (orthogonal) matrices is unitary (orthogonal) as well.

Proof Suppose U_1, U_2 are unitary (orthogonal), then

$$(U_1 U_2)^* U_1 U_2 = U_2^* U_1^* U_1 U_2 = U_2^* I U_2 = I.$$

From Lemma 6.2, we know the product is unitary (orthogonal).

Chapter 6. Structure of Operators in Inner Product Spaces

1.1. Use the upper triangular representations of an operator to give an alternative proof of the fact that the determinant is the product and the trace is the sum of eigenvalues counting multiplicities.

Proof (The proof uses the fact that the entries on the diagonal of T are the eigenvalues of A , counting multiplicity, which seems not to be explicitly stated in the book and can be found like in the Wiki.) $A = UTU^*$. $\det A = (\det U)(\det T)(\det U^*) = \det U = \prod_{i=1}^n \lambda_i$ because U is unitary and T is upper triangular with eigenvalues of A on its diagonal.

To consider the trace, suppose $U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n]$. Then A can be

represented by

$$\begin{aligned}
A &= [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_n] \begin{bmatrix} \lambda_1 & t_{12} & \dots & t_{1n} \\ & \lambda_2 & \dots & t_{2n} \\ & & \ddots & \vdots \\ 0 & & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \\ \vdots \\ \mathbf{u}_n^\top \end{bmatrix} \\
&= [\lambda_1 \mathbf{u}_1 \quad t_{12} \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 \quad \dots \quad t_{1n} \mathbf{u}_1 + t_{2n} \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n] \begin{bmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \\ \vdots \\ \mathbf{u}_n^\top \end{bmatrix} \\
&= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top + \dots + \lambda_n \mathbf{u}_n \mathbf{u}_n^\top,
\end{aligned}$$

where we exploit the orthogonality of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$. Then note that the trace of matrix $\mathbf{u}_i \mathbf{u}_i^\top = [u_{i1} \mathbf{u}_i \quad u_{i2} \mathbf{u}_i \quad \dots \quad u_{in} \mathbf{u}_i]$ (outer product) is $u_{i1}^2 + u_{i2}^2 + \dots + u_{in}^2 = \|\mathbf{u}_i\|^2 = 1$. Thus $\text{trace} A = \text{trace}(\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top) + \text{trace}(\lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top) + \dots + \text{trace}(\lambda_n \mathbf{u}_n \mathbf{u}_n^\top) = \lambda_1 + \lambda_2 + \dots + \lambda_n = \sum_{i=1}^n \lambda_i$.

2.2. True or false: The sum of normal operators is normal? Justify your conclusion.

Solution True. Suppose two normal operators are $N_1 = U_1 D_1 U_1^*, N_2 = U_2 D_2 U_2^*$. U_1, U_2 are unitary and D_1, D_2 are diagonal.

$$\begin{aligned}
(N_1 + N_2)^*(N_1 + N_2) &= N_1^* N_1 + N_1^* N_2 + N_2^* N_1 + N_2^* N_2 \\
(N_1 + N_2)(N_1 + N_2)^* &= N_1 N_1^* + N_1 N_2^* + N_2 N_1^* + N_2 N_2^*
\end{aligned}$$

N_1, N_2 are normal, we need to prove $N_1^* N_2 + N_2^* N_1 = N_1 N_2^* + N_2 N_1^*$. In fact, $N_1^* N_2 = U_1 D_1^* U_1^* U_2 D_2 U_2^*$. $N_1 N_2^* = U_1 D_1 U_1^* U_2 D_2^* U_2^*$. As can be shown, $D_1^* U_1^* U_2 D_2 = D_1 U_1^* U_2 D_2^*$ because D_1, D_2 are diagonal matrices, $D_1^* = \overline{D_1}, D_2^* = \overline{D_2}$. $D_1^* D_2 = D_1 D_2^*$ (for complex numbers $c_1, c_2, \overline{c_1} c_2 = c_1 \overline{c_2}$). By checking the entries of the product, one can conclude $D_1^* U_1^* U_2 D_2 = D_1 U_1^* U_2 D_2^*$ and $N_1^* N_2 = N_1 N_2^*, N_2^* N_1 = N_2 N_1^*$. So the statement is true.

2.9. Give a proof if the statement is true, or give a counterexample if it is false:

a) If $A = A^*$ then $A + iI$ is invertible.

True. The eigenvalues of $A + iI$ are $\lambda_i + i$ where λ_i are eigenvalues of A and are real. Then $\det(A + iI) = \prod_{i=1}^n (\lambda_i + i) \neq 0$. (If $c_1, c_2 \in \mathbb{C}, c_1 c_2 = 0$, then at least one of c_1, c_2 is 0.)

b) If U is unitary, $U + \frac{3}{4}I$ is invertible.

True. If $(U + \frac{3}{4}I)\mathbf{x} = U\mathbf{x} + \frac{3}{4}\mathbf{x} = \mathbf{0}$, note that $\|U\mathbf{x}\| = \|\mathbf{x}\|$. Then $\|U\mathbf{x} + \frac{3}{4}\mathbf{x}\| \geq \|U\mathbf{x}\| - \|\frac{3}{4}\mathbf{x}\| = \frac{1}{4}\|\mathbf{x}\|$. So the homogeneous equation only has the trivial solution, $U + \frac{3}{4}I$ is invertible.

- c) If a matrix is real, $A - iI$ is invertible.
False. A can have an eigenvalue i .

3.1. Show that the number of non-zero singular values of a matrix A coincides with its rank.

Proof Suppose the dimension of A is $m \times n$. It is known that $\text{Ker } A = \text{Ker } (A^*A)$. Then $\text{Rank } A = n - \dim \text{Ker } A = n - \dim \text{Ker } (A^*A) = \text{Rank } (A^*A)$. The SVD of A is $A = W\Sigma V^*$, $A^*A = V\Sigma^*W^*W\Sigma V^* = V\Sigma^2V^*$. $\text{Rank } A = \text{Rank } (A^*A) = \text{Rank } (V\Sigma^2V^*) = \text{Rank } \Sigma^2 = \text{number of non-zero singular values because } V \text{ is an orthogonal matrix (full rank)}.$

3.5. Find singular value decomposition of the matrix

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}.$$

Use it to find

- a) $\max_{\|\mathbf{x}\| \leq 1} \|A\mathbf{x}\|$ and the vector where the maximum is attained;
b) $\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$ and the vector where the minimum is attained;
c) the image $A(B)$ of the closed unit ball in $\mathbb{R}^2$, $B = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$. Describe $A(B)$ geometrically.

Solution (The SVD steps are ignored here.) a) Suppose $A = W\Sigma V^*$, then $(A\mathbf{x}, A\mathbf{x}) = \mathbf{x}^*A^*A\mathbf{x} = \mathbf{x}^*V\Sigma^2V^*\mathbf{x} = (V^*\mathbf{x})^*\Sigma^2(V^*\mathbf{x})$. Define $\mathbf{y} = [y_1 \ y_2]^T = V^*\mathbf{x}$. Because V is orthogonal, then $\mathbf{y}$ also lies in the unit ball. Thus

$$\begin{aligned} (A\mathbf{x}, A\mathbf{x}) &= \mathbf{y}^*\Sigma^2\mathbf{y} \\ &= [y_1 \ y_2] \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= 16y_1^2 + y_2^2. \end{aligned}$$

$y_1^2 + y_2^2 \leq 1$. Hence the maximum is 16 attained when $\mathbf{y} = [1 \ 0]^T$. Corresponding $\mathbf{x}$ can be solved by $\mathbf{x} = V\mathbf{y}$.

- b) Similarly, the minimum is 1 attained when $\mathbf{y} = [0 \ 1]^T$.
c) An ellipse.

3.8. Let A be an $m \times n$ matrix. Prove that *non-zero* eigenvalues of the matrices A^*A and AA^* (counting multiplicities) coincide.

Proof Suppose $\mathbf{v}$ is an eigenvector of A^*A corresponding to a non-zero eigenvalue λ , i.e., $A^*A\mathbf{v} = \lambda\mathbf{v}$. Then $AA^*A\mathbf{v} = A(\lambda\mathbf{v}) = \lambda(A\mathbf{v})$, i.e., λ is an eigenvalue of AA^* with corresponding eigenvector $A\mathbf{v}$. Similarly, we can show that non-zero eigenvalues of AA^* are also eigenvalues of A^*A . Thus

non-zero eigenvalues of the matrices A^*A and AA^* coincide.

4.2. Let A be a normal operator, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be its eigenvalues (counting multiplicities). Show that singular values of A are $|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|$.

Proof First, we show for normal operator A , $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$. Note that $AA^* = A^*A$, then

$$\begin{aligned} ((AA^* - A^*A)\mathbf{x}, \mathbf{x}) &= (\mathbf{0}, \mathbf{x}) \\ &= (AA^*\mathbf{x}, \mathbf{x}) - (A^*A\mathbf{x}, \mathbf{x}) \\ &= (A^*\mathbf{x}, A^*\mathbf{x}) - (A\mathbf{x}, A\mathbf{x}) \\ &= \|A^*\mathbf{x}\|^2 - \|A\mathbf{x}\|^2 \\ &= 0. \end{aligned}$$

Thus $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$.

Suppose $\mathbf{v}$ is an eigenvector of A corresponding to the eigenvalue λ . Note that $A - \lambda I$ is also normal (see Problem 2.2). Thus we have $\|(A - \lambda I)\mathbf{v}\| = \|(A - \lambda I)^*\mathbf{v}\| = \|(A^* - \bar{\lambda}I)\mathbf{v}\| = 0$, i.e., $A^*\mathbf{v} = \bar{\lambda}\mathbf{v}$. So $A^*A\mathbf{v} = A^*(\lambda\mathbf{v}) = \lambda\bar{\lambda}\mathbf{v} = |\lambda|^2\mathbf{v}$. $|\lambda|^2$ is an eigenvalue of A^*A , then $|\lambda|$ is a singular value of A .

4.4. Let $A = \widetilde{W}\widetilde{\Sigma}\widetilde{V}^*$ be a *reduced* singular value decomposition of A . Show that $\text{Ran } A = \text{Ran } \widetilde{W}$, and then by taking adjoint that $\text{Ran } A^* = \text{Ran } \widetilde{V}$.

Proof Suppose A is an $m \times n$ matrix. $\widetilde{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$. $\widetilde{W}$ is $m \times r$ and $\widetilde{V}^*$ is $r \times n$. To show $\text{Ran } A = \text{Ran } \widetilde{W}$, we just need to show $\text{Ran } \widetilde{\Sigma}\widetilde{V}^* = \mathbb{R}^r = \text{Ran } \widetilde{V}^*$ ($\widetilde{\Sigma}$ has full rank r), which holds because $\text{Rank } \widetilde{V}^* = r$ and $r \leq n$. Thus $\text{Ran } \widetilde{\Sigma}\widetilde{V}^* = \mathbb{R}^r$ and $\text{Ran } A = \text{Ran } \widetilde{W}$.

By taking adjoint, it can be easily shown that $\text{Ran } A^* = \text{Ran } \widetilde{V}$.

***** **End** *****